

Chapter One

Ordinary Differential Equations and their Solutions

Section-1.1

Basic Definitions and Preliminaries

1.1.2 Differential Equations

Definition: An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a differential equation.

Examples:

$$(i) \quad x dx + y dy = 0 \quad \text{or,} \quad \frac{dy}{dx} + \frac{x}{y} = 0 .$$

$$(ii) \quad \frac{d^2 y}{dx^2} + y \sin x = 0 .$$

$$(iii) \quad \frac{d^4 y}{dx^4} + x^2 \frac{d^3 y}{dx^3} + x^3 \frac{dy}{dx} = x e^x .$$

$$(iv) \quad \frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v .$$

$$(v) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 .$$

are differential equations, where in (i), (ii), and (iii) x is independent variable and y is dependent variable and in (iv) and (v) u and v are dependent variables and x, y, z, s and t are independent variables.

1.1.3 Types of Differential Equations

There are two types of differential equations. Such as:

- (i) Ordinary differential equations.
- (ii) Partial differential equations.

1.1.4 Ordinary Differential Equations

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

Example: $x dx + y dy = 0$ or, $\frac{dy}{dx} + \frac{x}{y} = 0$ is an ordinary differential equation, where x is independent variable and y is dependent variable.

1.1.5 Partial Differential Equation

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variables is called a partial differential equation.

Examples:

(i) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ is a partial differential equation, where u is dependent variable and x, y, z are independent variables.

(ii) $\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v$ is a partial differential equation, where v is dependent variable and s, t are independent variables.

1.1.6 Applications of Differential Equations

Differential equations are widely used in many disciplines together with engineering, mathematical and physical sciences, and other applied sciences that can be modeled by differential equations. There are many applications in real life problem. Some of them are as follows:

- (i) The study of the rate of growth or decay of a population.
- (ii) The study of the rate of disintegration of a radioactive substance.
- (iii) The study of the chemical reactions.
- (iv) The problem of determining the charge or current in an electric circuit.
- (v) The problem of determining the motion of a projectile, satellite, rocket or planet.
- (vi) The problem of determination of curves whose geometrical properties are known.
- (vii) The problem of determining the vibrations of a wire.
- (viii) The problem of the conducting of heat in a rod or in a slab.

1.1.7 Order of an Ordinary Differential Equation

The order of the highest ordered derivative involved in a differential equation is called the order of the differential equation.

Examples:

(i) $\frac{d^2 x}{dt^2} + 2\left(\frac{dx}{dt}\right)^3 + 7x = 0$ is an ordinary differential equation of order two.

(ii) $\left(\frac{d^2 y}{dx^2}\right)^3 + y = 0$ is an ordinary differential equation of order two.

(iii) $x^2 dy + y^2 dx = 0$ is an ordinary differential equation of order one.

(iv) $\frac{d^4 y}{dx^4} + 3\left(\frac{d^2 y}{dx^2}\right)^5 + 5y = 0$ is an ordinary differential equation of order four.

1.1.8 Degree of an Ordinary Differential Equation

The degree of the highest order differential of an ordinary differential equation is called the degree of an ordinary differential equation, where differential equation is polynomial differential equation of differential co-efficient.

Examples:

(i) $\frac{d^2x}{dt^2} + 2\left(\frac{dx}{dt}\right)^3 + 7x = 0$ is a first degree ordinary differential equation.

(ii) $\left(\frac{d^2y}{dx^2}\right)^3 + y = 0$ is a third degree ordinary differential equation.

(iii) $x^2 dy + y^2 dx = 0$ is a first degree ordinary differential equation.

(iv) $\frac{d^4y}{dx^4} + 3\left(\frac{d^2y}{dx^2}\right)^5 + 5y = 0$ is a first degree ordinary differential equation.

1.1.9 Linear Differential Equations

A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that is in, or can be expressed in the form

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \dots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = b(x),$$

where a_0 is not identically zero.

Or, A differential equation in which the dependent variable and its various derivatives occur to the first degree only, that no products of dependent variable and any of its derivatives are present and that no transcendental functions of dependent variable or its derivatives occur is called linear differential equation.

Examples:

(i) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$.

(ii) $\frac{d^4y}{dx^4} + x^2\frac{d^3y}{dx^3} + x^3\frac{dy}{dx} = xe^x$.

are both linear differential equations. Because in each case the dependent variable y and its various derivatives occur to the first degree only and that no products of y and any of its derivatives are present.

1.1.10 Non-Linear Differential Equations

A non-linear ordinary differential equation is an ordinary differential equation that is not linear.

Examples:

$$(i) \quad \frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y^2 = 0.$$

$$(ii) \quad \frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 + 6y = 0.$$

$$(iii) \quad \frac{d^2y}{dx^2} + 5y\frac{dy}{dx} + 6y = 0.$$

are both non-linear differential equations. Because in equation (i) the dependent variable y appears to the second degree in the term $6y^2$. Equation (ii) involves the third degree of the first derivative, i.e., non-linear term $5\left(\frac{dy}{dx}\right)^3$ present. And in equation (iii) the term $5y\frac{dy}{dx}$, which involves the product of the dependent variable and its first derivative.

1.1.11 Solution of a Differential Equation

If any define and differentiable function satisfies a differential equation, then the function is called a solution of the differential equation.

Examples:

$$(i) \quad y = \frac{x^2}{2} + C, \text{ where } C \text{ is arbitrary constant.}$$

$$(ii) \quad y = \frac{x^2}{2} + 2 \text{ are solutions of the differential equation } \frac{dy}{dx} = x.$$

1.1.12 General Solution of a Differential Equation

The relation containing n arbitrary constants which satisfy an ordinary differential equation of n -th order is called its complete primitive or general solution.

Examples:

$$(i) \quad y = \frac{x^2}{2} + C, \text{ is the general solution of the differential equation } \frac{dy}{dx} = x, \text{ where } C \text{ is arbitrary constant.}$$

$$(ii) \quad y = x^3 + Cx \text{ is not a general solution of differential equation } \frac{d^2y}{dx^2} = 6x. \text{ Because } \frac{d^2y}{dx^2} = 6x \text{ is a differential equation of order two. So it has two arbitrary constants.}$$

1.1.13 Particular Solution of a Differential Equation

A particular solution of a differential equation is one obtained from the primitive by assigning definite values to the arbitrary constants.

Example: $y = \frac{x^2}{2} + 1$ is an example of particular solution of differential equation $\frac{dy}{dx} = x$. Because, for

particular value of arbitrary constant $C = 1$, solution $y = \frac{x^2}{2} + C$ satisfy the differential equation $\frac{dy}{dx} = x$.

1.1.14 Singular Solution of a Differential Equation

A solution of a differential equation that contains no arbitrary constant and is not particular solution is called singular solution or singular integral.

Example: $y = 0$ is a singular solution of differential equation $(y')^2 = y$.

1.1.15 Explicit Solution of Differential Equation

A solution of a differential equation of the form $y = f(x)$ is called explicit solution of the differential equation.

Example: Solution of the differential equation $\frac{dy}{dx} = x$ is $y = \frac{x^2}{2} + C$, which is an example of explicit solution of the differential equation.

1.1.16 Implicit Solution of Differential Equation

Definition : A relation $f(x, y) = 0$ is said to be an implicit solution of a differential on the interval I if it defines one or more explicit solutions on I .

or, A solution of a differential equation of the form $f(x, y) = 0$ is called implicit solution of the differential equation.

Example: $x^2 - y^2 - c = 0$ is an implicit solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$.

Solved Problems

Example-1 Determine the order and degree of the following differential equations:

(i) $\frac{d^3 y}{dx^3} + 3\left(\frac{dy}{dx}\right)^4 + y = 0$. [NUH 2011]

(ii) $x\left(\frac{d^2 y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 - y = 0$. [NUH 2012]

(iii) $x\left[y\frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = y\frac{dy}{dx}$. [NUH(NM) 2006]

(iv) $\frac{d^2 y}{dx^2} = k\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{5}{3}}$.

$$(v) \left(\frac{d^2 y}{dx^2} + \frac{dy}{dx} \right)^2 = x \left(x \frac{d^2 y}{dx^2} + y \right)^{-3} . \quad [NUH 2010]$$

$$\text{Or, } (y'' + y')^2 = x(xy'' + y)^{-3} . \quad [JNUH 2021]$$

$$(vi) \sqrt[3]{\left(\frac{d^3 y}{dx^3} \right)^4 - 5x \frac{d^2 y}{dx^2} + y} = \sqrt[5]{\left(\frac{dy}{dx} \right)^2 + y^2 - x} . \quad [DU (7C) NM 2020; NUH(NM) 2008; JNUH 2021]$$

Solution

- (i) The given differential equation contains the highest third order differential coefficient. So its order is 3. Again since in this equation, the power of the highest ordered derivative is one. So the degree of the equation is 1.
- (ii) The given differential equation contains the highest second order differential coefficient. So its order is 2. Again since in this equation, the power of the highest ordered derivative is three. So the degree of the equation is 3.
- (iii) The given differential equation contains the highest second order differential coefficient. So its order is 2. Again since in this equation, the power of the highest ordered derivative is one. So the degree of the equation is 1.
- (iv) The given differential equation contains the highest second order differential coefficient. So its order is 2.

Again eliminating the fractional power of the equation we obtain the equation is of the form

$$\left(\frac{d^2 y}{dx^2} \right)^3 = k^3 \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^5$$

In this case, the power of the highest ordered derivative is 3. Hence the degree of the equation is 3.

- (v) The given differential equation contains the highest second order differential coefficient. So its order is 2.

Again eliminating the fractional power of the equation we obtain the equation is of the form

$$\left(\frac{d^2 y}{dx^2} + \frac{dy}{dx} \right)^2 \cdot \left(x \frac{d^2 y}{dx^2} + y \right)^3 = x$$

In this case, the power of the highest ordered derivative is 5. Hence the degree of the equation is 5.

- (vi) Eliminating the fractional power of the given equation we obtain the equation is of the form

$$\left[\left(\frac{d^3 y}{dx^3} \right)^4 - 5x \frac{d^2 y}{dx^2} + y \right]^5 = \left[\left(\frac{dy}{dx} \right)^2 + y^2 - x \right]^3$$

Since this equation involving highest third order differential coefficient. So its order is 3.

Again, the power of the highest ordered derivative is 20. Hence the degree of the equation is 20.

Example-2 Show that $y = (x^2 + c)e^{-x}$, where c is a constant, is the general solution of the differential equation $\frac{dy}{dx} + y = 2xe^{-x}$. Find a particular solution for the condition $y(-1) = e + 3$.

[DU (7C) 2022; NUH 1997, 1999, 2005, 2008; B. Sc. (Pass) 2012; JNUH 2018]

Solution

1st Part : Given equation is

$$y = (x^2 + c)e^{-x} \quad \dots (1)$$

$$\Rightarrow e^x y = x^2 + c$$

Differentiate with respect to x , we get

$$e^x \frac{dy}{dx} + e^x y = 2x$$

$$\Rightarrow \frac{dy}{dx} + y = 2x e^{-x}$$

$\therefore y = (x^2 + c)e^{-x}$ is the general solution of the differential equation $\frac{dy}{dx} + y = 2x e^{-x}$. [Showed]

2nd Part : Given condition,

$$y(-1) = e + 3$$

Putting $x = -1$ in equation (1), we get

$$y(-1) = (1 + c)e$$

$$\Rightarrow e + 3 = (1 + c)e$$

$$\Rightarrow 3 = ce$$

$$\therefore c = \frac{3}{e}$$

Putting value of c in equation (1), we obtain the particular solution

$$y = \left(x^2 + \frac{3}{e}\right)e^{-x}. \text{ (Ans.)}$$

Example-3 Show that, $v = \frac{A}{r} + B$ is a solution of the differential equation

$$\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0.$$

[B.Sc. (Pass) 2016]

Solution Try Yourself.

Example-4 Show that $y = cx + \frac{4}{c}$ is a general solution, where c is any arbitrary constant, $y = x + 4$ is a particular solution and $y^2 = 16x$ is a singular solution of the differential equation $x\left(\frac{dy}{dx}\right)^2 - y\left(\frac{dy}{dx}\right) + 4 = 0$.

Solution Try Yourself.

Example-5 Determine all such values of n for which the function defined by $y(x) = x^n$ is a solution of the differential equation

$$x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} - 10x \frac{dy}{dx} - 8y = 0 \quad . \quad [JNUH 2019]$$

Solution Given equation,

$$x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} - 10x \frac{dy}{dx} - 8y = 0 \quad \dots (1)$$

Since $y(x) = x^n$ is a solution of equation (1), we get

$$\begin{aligned} & x^3 \cdot n(n-1)(n-2)x^{n-3} + 2x^2 \cdot n(n-1)x^{n-2} - 10x \cdot nx^{n-1} - 8x^n = 0 \\ \Rightarrow & \{n(n-1)(n-2) + 2n(n-1) - 10n - 8\}x^n = 0 \\ \Rightarrow & n(n^2 - 3n + 2) + 2n^2 - 2n - 10n - 8 = 0 \quad [\because y(x) = x^n \neq 0] \\ \Rightarrow & n^3 - 3n^2 + 2n + 2n^2 - 12n - 8 = 0 \\ \Rightarrow & n^3 - n^2 - 10n - 8 = 0 \\ \Rightarrow & n^3 + n^2 - 2n^2 - 2n - 8n - 8 = 0 \\ \Rightarrow & n^2(n+1) - 2n(n+1) - 8(n+1) = 0 \\ \Rightarrow & (n+1)(n^2 - 2n - 8) = 0 \\ \Rightarrow & (n+1)(n^2 - 4n + 2n - 8) = 0 \\ \Rightarrow & (n+1)\{n(n-4) + 2(n-4)\} = 0 \\ \Rightarrow & (n+1)(n-4)(n+2) = 0 \\ \therefore & n = -1, -2, 4 \end{aligned}$$

(Ans.) $n = -1, -2, 4$.

Example-6 Show that, $x + y + e^{xy} = 0$ is an implicit solution to the nonlinear equation $(1 + xe^{xy})\frac{dy}{dx} + ye^{xy} + 1 = 0$.

Solution Given that, $x + y + e^{xy} = 0$... (1)

First, we observe that equation (1) is not solvable directly for y in terms of x alone. Now, using the technique of implicit differentiation, differentiating (1) w. r. to x , we get

$$1 + \frac{dy}{dx} + e^{xy} \left(y + x \frac{dy}{dx} \right) = 0$$

$$\Rightarrow (1 + xe^{xy}) \frac{dy}{dx} + ye^{xy} + 1 = 0$$

which is identical to the given differential equation.

Thus, the relation (1) is an implicit solution to the given differential equation. **[Showed]**

Example-7

(i) Verify that the expression $-2x^2y + y^2 = 1$ is an implicit solution of the first order differential equation $2xy dx + (x^2 - y)dy = 0$. **[JNUH 2022]**

(ii) Find at least one explicit solution $y = \phi(x)$. Give an interval I of each solution $\phi(x)$.

Solution (i) Given expression, $-2x^2y + y^2 = 1$... (1)

First, we observe that (1) is not solvable directly for y in terms of x alone. By implicit differentiation, differentiating (1) w. r. to x , we get

$$-2x^2 \frac{dy}{dx} - 4xy + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow -4xy - 2(x^2 - y) \frac{dy}{dx} = 0$$

$$\therefore 2xy dx + (x^2 - y)dy = 0$$

which is identical to the given equation $2xy dx + (x^2 - y)dy = 0$.

Thus, the relation (1) is an implicit solution to the given differential equation. **[Verified]**

(ii) From (1), we get

$$y^2 - 2x^2y - 1 = 0$$

$$\Rightarrow y = \frac{-(-2x^2) \pm \sqrt{4x^4 + 4}}{2 \cdot 1}$$

$$\Rightarrow y = x^2 \pm \sqrt{x^4 + 1}$$

∴ The explicit solutions are : $\phi_1(x) = x^2 + \sqrt{x^4 + 1}$ and

$$\phi_2(x) = x^2 - \sqrt{x^4 + 1}$$

The above two explicit solutions are valid in the interval $(-\infty, \infty)$. (**Ans.**)

Example-8 Verify that $3x^2 - y^2 = c$ is a one-parameter family of solutions of the differential equation $y \frac{dy}{dx} = 3x$. Find all explicit solutions $y = \phi(x)$ of the differential equation. Give the interval I of definition of each explicit solution. Which of the explicit solutions satisfies $y(-2) = 3$? [**DUH 2019, 2021**]

Solution Try Yourself.

Exercise - 1.1

1. Define differential equation, order and degree of an ordinary differential equation with example.

[**Ans. Section 1.1.2, 1.1.7, 1.1.8**]

2. Define linear and non-linear differential equation or, Differentiate between linear and non-linear differential equations with example.

[**Ans: Section 1.1.9, 1.1.10**]

3. Classify each of the following differential equations as ordinary or partial differential equations, state the order of each equation; and determine whether the equation under consideration is linear or nonlinear.

(i) $\frac{dy}{dx} + x^2y = xe^x$ (ix) $\frac{d^2y}{dx^2} + x \sin y = 0$

(ii) $x^2dy + y^2dx = 0$ (x) $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$

(iii) $\frac{d^2y}{dx^2} + y \sin x = 0$ (xi) $\frac{d^2x}{dt^2} + k^2x = 0$

(iv) $\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 3y = \sin x$ (xii) $yy'' = x$

(xiii) $(x^2 + y^2)dx + 2xy dy = 0$

(v) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ (xiv) $y' + P(x)y = Q(x)$

(xv) $y''' - 3y' + 2y = 0$

(vi) $\frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + u = 0$

(xvi) $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

[**JNUH 2022**]

(xvii) $(\sin \theta)y''' - \cos \theta y' = 2$

[**JNUH 2022**]

$$(vii) \frac{d^4 y}{dx^4} + 3 \left(\frac{d^2 y}{dx^2} \right)^5 + 5y = 0$$

$$(xviii) \ddot{x} - \left(1 - \frac{\dot{x}^2}{3} \right) \dot{x} + x = 0$$

[JNUH 2022]

$$(viii) \frac{d^6 x}{dt^6} + \left(\frac{d^4 x}{dt^4} \right) \left(\frac{d^3 x}{dt^3} \right) + x = t$$

4. Write down some applications of differential equations. [Ans. Section 1.1.6]
5. What do you mean by solution of a differential equation? [Ans. Section 1.1.11]
6. Show that every function f defined by $f(x) = 2 + ce^{-2x^2}$, where c is an arbitrary constant, is a solution of the differential equation $\frac{dy}{dx} + 4xy = 8x$.
7. For certain value of the constant r the function f defined by $f(x) = e^{rx}$ is a solution of the differential equation

$$\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 12y = 0. \quad [\text{Ans. } 2, 3, -2]$$

Determine all such values of r .

8. Show that $x^3 + 3xy^2 = 1$ is an implicit solution of the differential equation $2xy \frac{dy}{dx} + x^2 + y^2 = 0$ on the interval $0 < x < 1$.
 9. Show that $5x^2y^2 - 2x^3y^2 = 1$ is an implicit solution of the differential equation $x \frac{dy}{dx} + y = x^3y^3$ on the interval $0 < x < \frac{5}{2}$.
 10. Show that every function f defined by $f(x) = c_1 e^{4x} + c_2 e^{-2x}$, where c_1 and c_2 are arbitrary constants, is a solution of the differential equation
- $$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 8y = 0.$$
11. Show that the function f defined by $f(x) = (2x^2 + 2e^{3x} + 3)e^{-2x}$ satisfies the differential equation
- $$\frac{dy}{dx} + 2y = 6e^x + 4xe^{-2x}$$
- and also the condition $f(0) = 5$.

The End